PACLAND · Physics, in plain words

The string landscape: what we checked, what we found, and what it means. One page, no jargon.

1 · The question

String theory says the universe could have about 10^500 different shapes — far more than there are atoms in the part of the universe we can see. Each shape would give different particle masses. So how would you ever tell which one is ours? The usual answer is: you can't. That is the question we took seriously.

2 · What we did

We stopped asking the theory for its answer and asked what it forbids. Every shape has to obey one rule: the total amount of flux — a kind of electric charge smeared across hidden dimensions — cannot exceed a limit that the shape itself sets. That limit is a single number, and it is exact. No measurement needed, no approximation.

3 · What we found

- There is exactly ONE exact number, and it is a total. Picture a household budget: the total spend is fixed, but the individual receipts are not. Here the total flux is exact, and how it is split between the hidden dimensions is completely free.
- That is provable, not a hunch. A theorem says every quantity that survives the theory's symmetries is a power of that one number — so there is nothing else exact to find. We machine-checked the proof with a computer, step by step. It is not “we believe”; it is “the computer verified it”.
- The 19 free numbers of particle physics stay free. The electron's mass, the strength of the forces — nobody has ever derived those values without measuring them, and this work says why: a final theory can supply relations between numbers, not the numbers.
- So one sentence is always wrong. “The theory predicts the electron mass is X” is either a fit in disguise or a mistake. The correct sentence is: the theory fixes the rules and the counts, and leaves the values free.

4 · How we score it

We count what can be checked exactly: the labels the shape carries (its topology), the one constraint (the total flux), and the exact counts (the 19 free parameters). That count is nine today, and every rung is machine-checked rather than argued. A claim that cannot be checked is not counted at all — which is the whole discipline in one line.

5 · What it means

A theory of everything predicts relations, never values. That sounds like a weakness and is actually why it can be tested: relations can be checked against a table, values can only be checked against an experiment you may never be able to run. So: don't build a detector to measure a number the theory cannot give you, and don't believe a claim that a theory has derived one.

6 · What this cannot do

It cannot tell you which vacuum is ours, or what the masses are. Nobody can, yet. What it can do is say precisely where the free choices are — and that is the part that can be checked, which is more than the field had before.